Connectome-Constrained Digital Twins of Longitudinal Neurodegeneration on OASIS-2: Network Diffusion, Graph Neural ODE and Generative Forecasting with an Interactive 3D Simulator

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***Abstract*—We build a regional *digital twin* pipeline on the longitudinal OASIS-2 cohort (150 older adults, 373 MRI sessions) to estimate which brain regions show atrophy and which are likely to follow. Each subject lives on a 56-region graph. We convert regional volumes to age-normalised atrophy scores (W-scores) and propagate them over a connectome Laplacian. Four forecasters run on this representation: the linear Network Diffusion Model (NDM) as a baseline, and three learned models built for irregularly spaced visits, a spatio-temporal Graph Neural ODE (ST-GNN-ODE), a physics-regularised Network-Diffusion VAE (ND-VAE) and a temporally conditioned diffusion model (TADM). A region-to-domain weight map converts regional atrophy into eleven cognitive-domain deficit scores, and an interactive 3D simulator shows the result with a time slider, region focus and an explainability panel. We analyse two simulator sessions, a nondemented control and a subject with CDR 0.5. In the second, large fitted diffusivities on our surrogate connectome flatten the NDM state, which matches the identical domain scores at year 3.3. We report no predictive benchmarks.**

***Index Terms*—digital twin, Alzheimer's disease, network diffusion model, neural ODE, graph neural network, longitudinal MRI, OASIS-2**

I. Introduction

Neurodegenerative disease advances along anatomically structured paths: atrophy typically appears first in medial temporal regions and spreads through connected cortex. For any patient we want to know which regions are affected now and which are likely to be affected next. Answering both from routine structural MRI would support monitoring and counselling without PET or fluid biomarkers.

The Network Diffusion Model (NDM) of Raj et al. [1], [2] explains spread as linear diffusion of pathology over the structural connectome, $x(t) = e^{-\beta L t} x(0)$. It is compact and interpretable, but it is linear, has a single scalar rate, and cannot express region-specific dynamics. Longitudinal cohorts such as OASIS-2 [3] add a further difficulty: follow-up intervals are irregular (a few months to over seven years), which discrete-time recurrent models handle poorly.

We build a pipeline that treats the NDM as a physics-based baseline, add three learned continuous-time forecasters on the same 56-region graph, and show the results in an interactive 3D digital-twin simulator. Our contributions are:

1) An end-to-end OASIS-2 pipeline from segmentation to age-normalised regional atrophy scores, a connectome Laplacian, and a per-subject NDM fit (Sections III–IV).
2) Implementations of ST-GNN-ODE, ND-VAE and TADM that share one dataset, one graph and one evaluation harness, with subject-level data splits.
3) A region-to-cognitive-domain deficit mapping that turns a regional atrophy vector into eleven domain scores.
4) A qualitative analysis of two simulator case studies, including a diagnosis of an over-smoothing behaviour of the diffusion baseline (Section V).

II. Related Work

Network diffusion and spread models. Raj et al. introduced the NDM for dementia [1] and showed that, with a healthy reference connectome, it predicts longitudinal atrophy and metabolism patterns in Alzheimer's disease [2]. We follow this "reference connectome" practice because OASIS-2 contains no diffusion MRI.

Continuous-time and graph models. Neural ODEs [4] define hidden-state dynamics by a learned vector field and integrate to arbitrary times; latent-ODE variants target irregularly sampled series [5]. We use graph attention [6] and graph convolution [7] as the spatial operator. From physics-informed learning [8] we take the idea of penalising deviation from a governing equation and apply it to the diffusion law.

Generative forecasting. We use a variational autoencoder [9] to obtain a probabilistic latent trajectory code. Our conditional

generator builds on denoising diffusion models [10], their fast implicit sampler [11], the cosine noise schedule [12] and feature-wise conditioning [13].

Digital twins and evaluation practice. Virtual brain twins personalise whole-brain network models to individuals [14]. For MRI classification, slice-level splitting leaks information and inflates accuracy; on OASIS the reported inflation is about 30 percentage points [15], and reproducible evaluation protocols are discussed in [16]. We therefore split by subject. The simulator's explainability note refers to integrated gradients [17].

III. Data and Preprocessing

A. Cohort

OASIS-2 contains longitudinal T1-weighted MRI from 150 subjects aged 60–96 acquired on one scanner [3]. The demographics file in our repository lists 373 sessions: 94 subjects have two visits, 43 have three, 9 have four and 4 have five (Fig. 1). At baseline 72 subjects are nondemented, 64 demented and 14 "converted" (nondemented at entry, demented later); 88 are female and 62 male. Baseline age is 75.4 ± 7.5 years and baseline MMSE 27.6 ± 3.0. Across all sessions the Clinical Dementia Rating (CDR) is 0 in 206, 0.5 in 123, 1 in 41 and 2 in 3. The longest follow-up from baseline is 2639 days (7.2 years). The learned models read the tabular covariates and clinical targets directly; their node features come from the segmentation step below.

B. Regional parcellation and W-scores

The pipeline defines $N = 56$ regions named in the Desikan–Killiany convention [18]: 34 left-hemisphere cortical labels, 10 subcortical structures (bilateral hippocampus, amygdala, thalamus, caudate and putamen), and 12 right-hemisphere cortical labels. We obtain regional volumes with SynthSeg [19] where it is installed and with an FSL-based fallback otherwise, and normalise them by estimated total intracranial volume (eTIV).

To remove normal ageing from the pathology signal we compute W-scores. For each region we fit a linear regression of normalised volume on age using sessions with CDR $= 0$, standardise each session's residual by the residual mean and standard deviation of that reference set, and flip the sign so that atrophy is positive. A W-score of $+1$ means one standard deviation less tissue than expected for that age. The resulting vector $x \in \mathbb{R}^{56}$ is the node-feature channel of every model.

C. Template connectome and Laplacian

We constrain regional spread with a 56×56 template adjacency A. The shipped template is a *synthetic surrogate*: we generated it with a fixed seed from exponentially distributed weights, thresholded at the 80th percentile, and added edge weight (+0.5) among hippocampus, entorhinal and parahippocampal regions to encode medial-temporal vulnerability. The matrix is symmetric with 20.8% non-zero entries. The Laplacian is $L = D - A$ with D the weighted degree matrix; its eigenvalues range from 0 to 15.9, with the smallest non-zero value $\lambda_2 = 2.58$ (Fig. 3a). A connectome measured on real subjects (for example the Budapest or HCP atlases) should replace this surrogate; see Section VI.

IV. Methods

A. NDM baseline

Pathology spreads as $\dot{x} = -\beta L x$, with closed form $x(t) = e^{-\beta L t} x(0)$. With the eigendecomposition $L = V \Lambda V^{\top}$ computed once,

$$x(t) = V \,\mathrm{diag}\big(e^{-\beta \lambda_i t}\big)\, V^{\top} x(0). \tag{1}$$

For each subject we fit the scalar β by bounded one-dimensional minimisation of the squared error between predicted and observed W-scores at the first follow-up, with $\beta \in [0.001, 10]$ and t in years. For validation we use subjects with at least three visits: we fit β on visit 2, predict visit 3, and compute Pearson correlation and MSE over regions.

B. Learned forecasters

All three models take the baseline state $X(t_0) \in \mathbb{R}^{N \times 1}$, static covariates $c \in \mathbb{R}^6$ (age, sex, education, SES, eTIV, ASF, each scaled), the normalised adjacency $\hat{A} = \tilde{D}^{-1/2}(A + I)\tilde{D}^{-1/2}$ with $\tilde{D}$ the degree matrix of $A + I$, and irregular visit times in years from baseline (Fig. 4). Clinical heads predict MMSE (scaled to $[0, 1]$) and a four-class CDR stage (0, 0.5, 1, 2).

1) ST-GNN-ODE: An MLP encodes $[X(t_0) \,\|\, c]$ to $h(t_0) \in \mathbb{R}^{N \times 32}$. The latent state follows

$$\frac{dh}{dt} = f_\theta(h, t) = \mathrm{MLP}\big(\mathrm{GAT}(h; \hat{A}) \,\|\, c \,\|\, \phi(t)\big), \tag{2}$$

where GAT is a single-head graph attention layer [6] masked by $\hat{A}$ and $\phi(t)$ is a 16-dimensional sinusoidal time embedding. We advance the state to each visit time with five explicit Euler steps. An adaptive solver [4] would also work, but fixed steps run faster. A linear head reconstructs $\hat{X}(t_k)$ and two small heads on the mean-pooled state predict MMSE and CDR. The loss is $\mathrm{MSE}(\hat{X}, X) + 0.5\,\mathrm{MSE}_{\mathrm{MMSE}} + 0.3\,\mathrm{CE}_{\mathrm{CDR}}$.

2) ND-VAE: A GRU reads the visit sequence $[X(t_k) \,\|\, c]$ and outputs $(\mu, \log \sigma^2)$ of a 16-dimensional latent code z. A decoder maps $(z, t, c) \mapsto \hat{X}(t)$. The objective is

$$\begin{aligned}\mathcal{L} = \|\hat{X} - X\|_1 + \beta_{\mathrm{KL}}\, D_{\mathrm{KL}} + \lambda_{\mathrm{phys}} \left\| \tfrac{d\hat{X}}{dt} + \beta_{\mathrm{d}} L \hat{X} \right\|^2 \\ + 0.5\,\mathrm{MSE}_{\mathrm{MMSE}} + 0.3\,\mathrm{CE}_{\mathrm{CDR}}.\end{aligned} \tag{3}$$

We anneal β_{KL} linearly to 0.1 over 20 epochs, set $\lambda_{\mathrm{phys}} = 0.01$, initialise a learnable diffusion coefficient β_{d} at 0.1, and approximate the time derivative by finite differences between consecutive visits. Here $L = I - \hat{A}$.

3) TADM: For each baseline–future pair, we noise the target X_k with a cosine schedule [12] over $T = 200$ steps. A conditioning vector e_{cond} concatenates a pooled graph-convolution [7] embedding of $X(t_0)$, a sinusoidal embedding of Δt_k, and a covariate embedding, projected to 32 dimensions. An MLP denoiser with FiLM modulation [13] predicts the added noise, and training minimises $\|\epsilon - \epsilon_\theta(X_k^{(\tau)}, \tau, e_{\mathrm{cond}})\|^2$ [10]. Sampling uses 10 DDIM steps [11]; repeated sampling from different noise seeds gives a spread that we use as an uncertainty estimate.

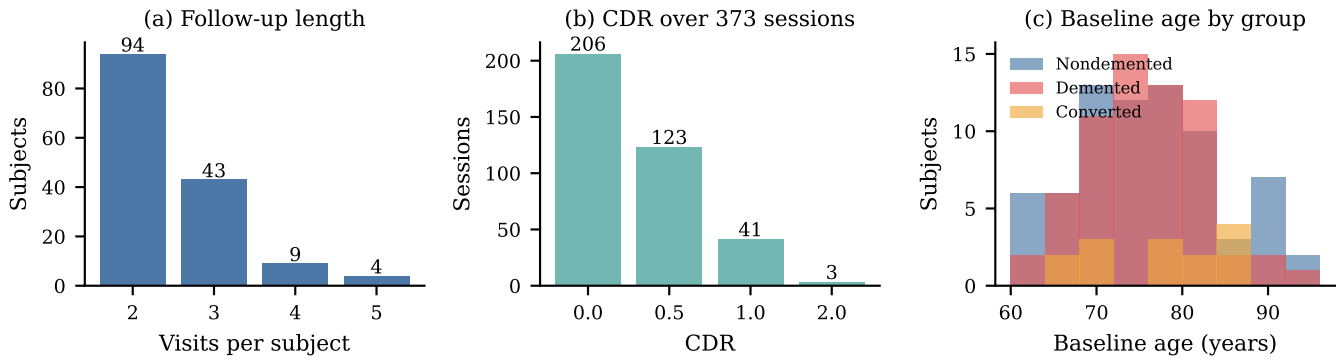


Fig. 1. OASIS-2 cohort as used here (from the demographics file). (a) Number of visits per subject. (b) CDR distribution over all 373 sessions. (c) Baseline age by baseline group.

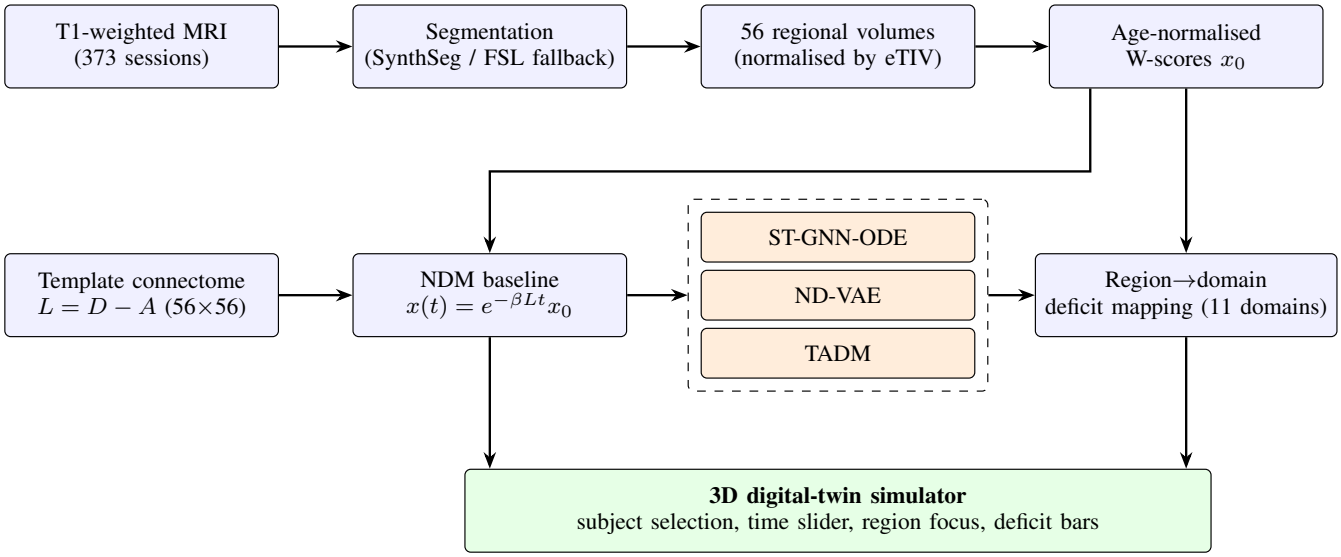


Fig. 2. Pipeline overview. Segmentation yields regional volumes that are normalised to W-scores; the template connectome supplies the Laplacian used by the NDM baseline and the graph operators of the learned forecasters; predicted regional atrophy is mapped to cognitive-domain deficits and displayed in the 3D simulator.

C. Training protocol and metrics

We split subjects with at least two visits into 70/15/15% train/validation/test sets (seed 42) and keep all of a subject's visits in one set, following the leakage-aware practice of [15]. We train with AdamW (learning rate 10^{-3}, weight decay 10^{-4}), cosine learning-rate decay, gradient clipping at 1.0, batch size 8, and early stopping on validation loss (patience 10, 10 and 15; at most 50, 100 and 200 epochs for ST-GNN-ODE, ND-VAE and TADM). The evaluation harness reports atrophy RMSE, MAE and Pearson r, MMSE MAE, and CDR accuracy and macro-F1, alongside the NDM baseline. The numerical benchmark is outside the scope of this paper.

D. Cognitive-domain deficit mapping

A weight map $w_{r,d} \in [0, 1]$ links regions to eleven domains (episodic memory, executive function, language, visuospatial, emotional regulation, attention, working memory, semantic memory, learning, object recognition, behavioural control). For example, left hippocampus carries weight 1.0 for episodic memory and 0.8 for learning, and the left rostral middle frontal gyrus carries 1.0 for executive function. The deficit of domain d is the weight-averaged atrophy $s_d = \sum_r w_{r,d} x_r / \sum_r w_{r,d}$ over regions with non-zero weight. The implementation then min–max normalises the eleven scores to $[0, 1]$ within the subject whenever their range is non-zero, so each score ranks a domain within one subject and does not estimate risk.

E. Interactive 3D simulator

The simulator is the user-facing layer of the twin. From its interface (Figs. 5 and 6) it offers: (i) a subject selector showing clinical group, age/sex, baseline CDR and the subject's fitted NDM diffusivity β; (ii) camera presets (superior, anterior, left lateral, isometric) with toggles for connectome tracts and auto-rotation; (iii) a 3D view of 56 region nodes coloured by atrophy W-score on a scale from healthy (≤ 0) to severe ($\geq +3$); (iv) a "timeline horizon" slider with a Simulate/Pause

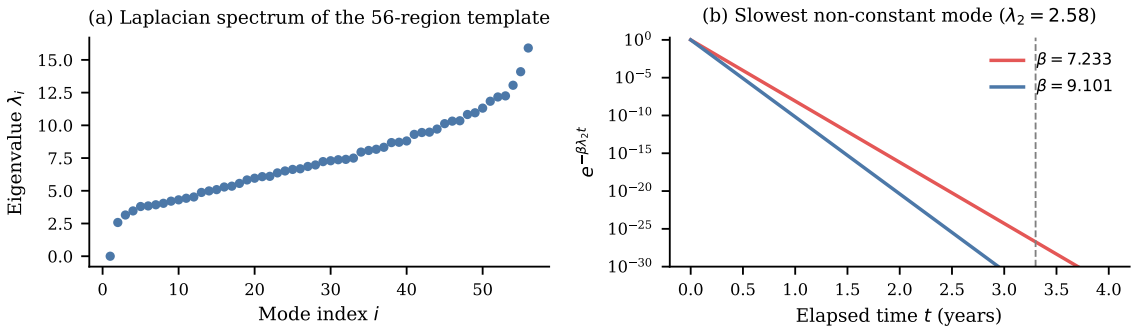


Fig. 3. Spectral view of the template connectome. (a) Laplacian eigenvalues of the 56-region surrogate. (b) Attenuation $e^{-\beta\lambda_2 t}$ of the slowest non-constant diffusion mode for the two diffusivities shown in the simulator screenshots; the dashed line marks $t = 3.3$ years. The y-axis is logarithmic.

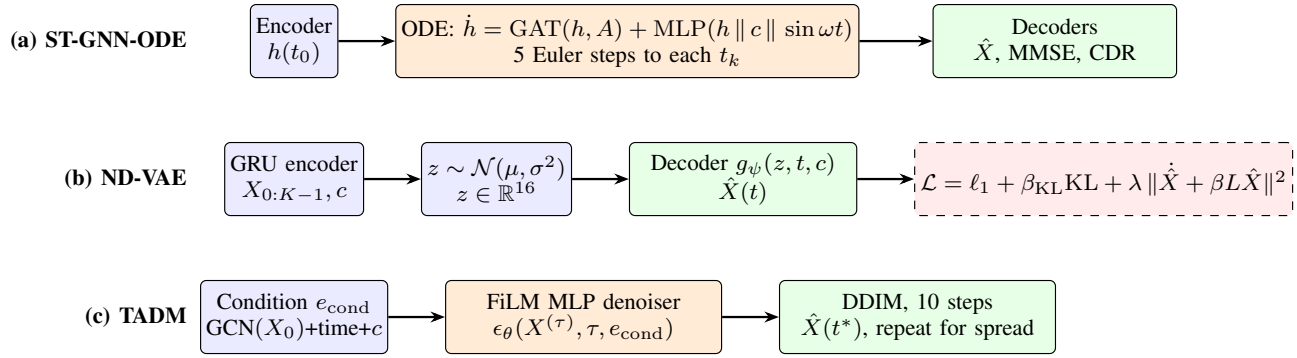


Fig. 4. The three learned forecasters as implemented. (a) A graph-attention vector field integrated continuously in time. (b) A GRU-encoded latent trajectory code with a diffusion-physics penalty. (c) A conditional denoising generator sampled with DDIM.

control, annotated with the propagation law $e^{-\beta L t}x_0$; and (v) a side panel with the focused region, its simulated atrophy in σ units, the eleven domain deficit bars, and an explainability note. We describe only what the interface shows.

V. Results: Simulator Case Studies

We analyse two simulator sessions, one per figure, and read all values below from the screenshots.

A. Case A: nondemented control at baseline

Fig. 5 shows subject OAS2_0001 (nondemented, 87 years, male, baseline CDR 0.0, fitted $\beta = 9.101$) at timeline year 0.0, before any propagation. The 3D view shows 56 nodes of different colour: most are cyan to blue, indicating W-scores near or below the healthy end of the scale, and a few are green, indicating mildly elevated atrophy. The focused region is the left insula (a left-hemisphere cortical parcel) with simulated atrophy $+0.82\,\sigma$, associated with emotional regulation and interoception.

The domain panel shows a heterogeneous profile: object recognition (73%), emotional regulation (71%), behavioural control (66%), attention and semantic memory (both 61%), episodic memory (55%), working memory (48%), learning and visuospatial (both 43%), executive function (40%) and language (35%). The ordering follows which weighted regions carry the highest W-scores. Because we min–max normalise scores within a subject (Section IV), 73% marks this subject's most affected domain and carries no risk meaning; the subject has CDR 0.0. The "Severe Pathology" label on the colour bar names the scale's maximum and says nothing about this subject.

B. Case B: subject with CDR 0.5 at year 3.3

Fig. 6 shows subject OAS2_0002 (labelled demented, 75 years, male, baseline CDR 0.5, fitted $\beta = 7.233$) while the simulation is running at timeline year 3.3 (the control reads "Pause"). The focused region is the left hippocampus, a subcortical nucleus in the left hemisphere, with simulated atrophy $+0.94\,\sigma$ and associations with episodic memory and learning. The hippocampus is among the structures most closely tied to early Alzheimer's pathology. The explainability note names the left hippocampus and entorhinal cortex as the primary baseline network drivers of longitudinal spread. Our surrogate connectome gives these regions extra edge weight, so this agreement holds by construction and offers no independent support for the attribution.

All eleven domain bars read an identical 94%, and the 3D nodes have a uniform colour. The displayed regional value ($+0.94\sigma$) also equals the domain value, as it should if every region holds the same value. The diffusion law predicts this. Under $e^{-\beta L t}$ every non-constant mode decays and the state converges to the regional mean of x_0. On the surrogate

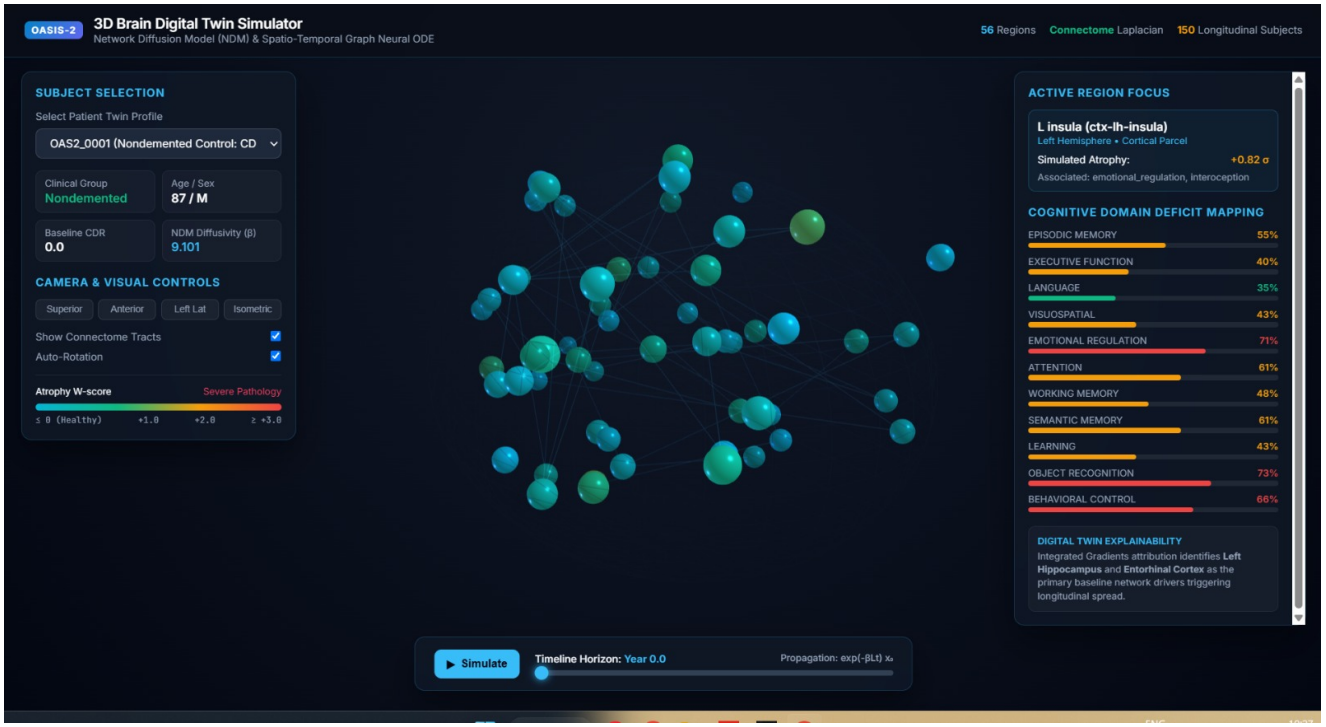


Fig. 5. Simulator view, Case A: subject OAS2_0001 (nondemented, 87 y, male, CDR 0.0, $\beta = 9.101$) at year 0.0. Left: subject selection, camera controls and atrophy W-score legend. Centre: 56-region 3D view with connectome tracts. Right: focused region (left insula, $+0.82\sigma$), eleven cognitive-domain deficit bars and the explainability note. Bottom: Simulate control and timeline slider.

connectome the slowest such mode has $\lambda_2 = 2.58$ (Fig. 3a), so the fitted $\beta = 7.233$ shrinks it by a factor of about 9×10^{-5} after only 0.5 years and about 2×10^{-27} at 3.3 years (Fig. 3b); for Case A's $\beta = 9.101$ the factors are 8×10^{-6} and 2×10^{-34}. If the simulator measures t in years, as the β-fitting routine does, the NDM state has flattened to its mean by year 3.3. The deficit map averages regional values, so a constant state gives the same score in every domain and produces the uniform bars. We could not inspect the simulator code, so this explanation remains a hypothesis. At diffusivities this large the baseline NDM keeps almost no spatial information beyond the first few months, and the saturated view says nothing about clinical severity.

C. Comparison of the two cases

Table I contrasts the two sessions. The control at baseline shows a differentiated profile with a low-grade insular focus, while the CDR 0.5 subject at year 3.3 shows a hippocampal focus of larger magnitude and a flattened domain profile. The fitted diffusivities also differ ($\beta = 9.101$ versus 7.233), and both lie in the upper part of the allowed range $[0.001, 10]$; a fit near the bound may mean the data constrain β only weakly. The explainability text is identical in both sessions although the focused regions differ, so we read it as a population-level statement about baseline drivers.

TABLE I
SUMMARY OF THE TWO SIMULATOR SESSIONS (VALUES READ FROM THE SCREENSHOTS).

	Case A	Case B
Subject	OAS2_0001	OAS2_0002
Group (label)	Nondemented	Demented
Age / sex	87 / M	75 / M
Baseline CDR	0.0	0.5
NDM β	9.101	7.233
Timeline	Year 0.0	Year 3.3
Focused region	L insula	L hippocampus
Simulated atrophy	$+0.82\sigma$	$+0.94\sigma$
Domain bars (range)	35–73%	94% (all 11)

VI. Discussion and Limitations

What the twin shows. The simulator shows a subject's current regional burden (Case A), a focus region with a plausible cognitive association (both cases) and a deficit profile across eleven domains, and it lets the user move through time. The hippocampal focus for the CDR 0.5 subject agrees with the established medial-temporal onset of Alzheimer's pathology.

Limitations.

1) *Surrogate connectome.* The template is synthetic. Spread patterns and the fitted β reflect its construction, and its density and large eigenvalue gap probably worsen the over-smoothing in Case B.
2) *No predictive benchmark.* The harness reports RMSE, MAE, Pearson r, MMSE MAE and CDR accuracy and F1, but we do not report them, so the qualitative results support no claim of forecasting accuracy.
3) *Simplified models.* We implemented lighter models than

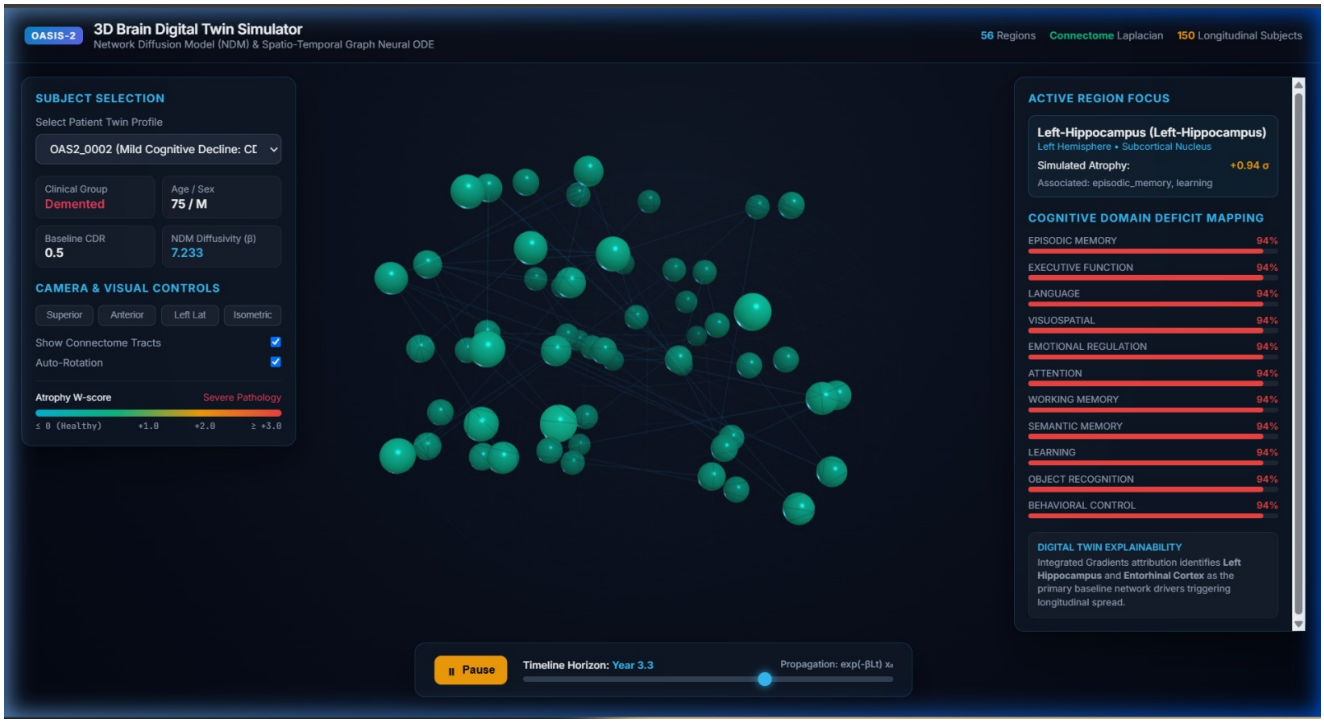


Fig. 6. Simulator view, Case B: subject OAS2_0002 (demented, 75 y, male, CDR 0.5, $\beta = 7.233$) at year 3.3 while simulating. The focused region is the left hippocampus ($+0.94\sigma$). All eleven domain deficit bars show 94% and the nodes are uniformly coloured, consistent with a spatially homogeneous state.

the design specifications call for: single-head attention with fixed Euler steps instead of an adaptive solver, a finite-difference physics loss instead of automatic differentiation, and a small MLP denoiser.

4) *Single feature channel.* Node features are one W-score per region. The dataset builder derives later-visit features from the baseline W-score with an exponential time modulation instead of from independently measured follow-up volumes, so reconstruction metrics on that data say nothing about forecasting skill.
5) *Relative deficit scores.* We rescale domain bars within a subject, so they cannot be read as absolute risk, and a flat state removes the contrast.
6) *Small, single-site cohort* (150 subjects) with no molecular or functional data.
7) *Simulator.* We could not inspect its source, so the Case B explanation remains a hypothesis.

VII. Conclusion and Future Work

We built a connectome-constrained digital-twin pipeline for OASIS-2 that combines a linear diffusion baseline, three continuous-time learned forecasters, a region-to-domain deficit mapping and an interactive 3D simulator, and we analysed two simulator sessions. The control at baseline showed a heterogeneous domain profile. The CDR 0.5 subject at year 3.3 showed a flattened one, which we attribute to large diffusivities on the surrogate connectome. Future work has five parts: replace the surrogate with a measured reference connectome; run the benchmark with leakage-free subject-level cross-validation, including three-visit hold-out forecasting; bound or regularise β and replace the within-subject min–max rescaling with a calibrated score; evaluate the TADM uncertainty spread; and run a usability study with clinicians.

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